4-Bit Binary Adder-Subtractor Specification

1. Circuit Overview

A 4-bit adder-subtractor is a fundamental arithmetic circuit used in CPUs to perform addition and subtraction. It uses XOR gates to conditionally invert the second operand for subtraction (2's Complement).

2. Full Adder Truth Table

A	B	Cin	Sum (S)	Carry (Cout)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

3. K-Maps

K-Map for Sum (S)

A \ BCin	00	01	11	10
0	0	1	0	1
1	1	0	1	0

K-Map for Carry (Cout)

A \ BCin	00	01	11	10
0	0	0	1	0
1	0	1	1	1

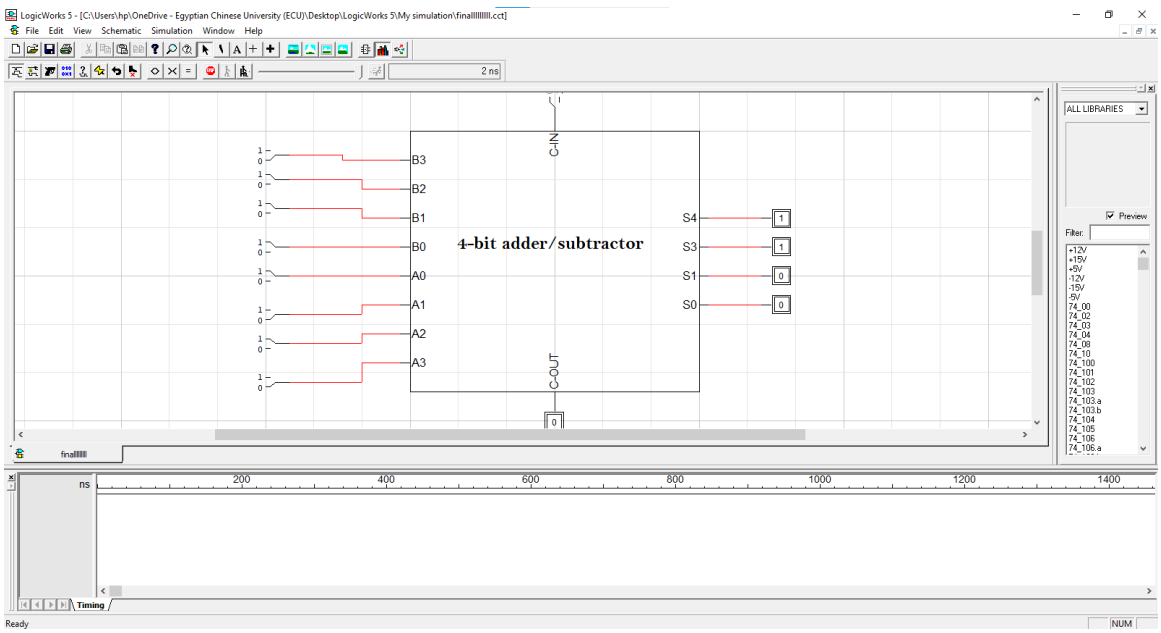
4. Simplified Logic Equations

Based on the K-Map minimization:

$Sum = A \oplus B \oplus Cin$

$Cout = AB + Cin(A \oplus B)$

1-ADDITION:-



2-SUBTRACYION:-

